Optimality conditions and pivot rules

Outline

Sufficient conditions for optimality

Necessary conditions for optimality
- Degenerate basic feasible solutions

Pivot rules
- Anti-cycling pivot rules
  - Bland's pivot rule

Sufficient conditions for optimality

Introduction: Sufficient conditions for optimality

- The simplex algorithm moves from a basis to another one: when can it **stop**?
- In this section we prove that a basis with $\bar{\boldsymbol{b}} \geq 0$ and $\bar{\boldsymbol{c}} \leq 0$ gives an **optimal solution**, and that with $\bar{\boldsymbol{c}} < 0$ this solution is the **unique** optimal one
- We also see that these conditions are only **sufficient**: an example shows an optimal basic solution with a positive reduced cost
- These results justify the stopping criterion of the simplex algorithm

The LP problem in standard form

- We consider an LP problem in **standard maximization form** with $n + m$ non-negative variables and m equality linear constraints:

$$\max \quad \boldsymbol{c}' \boldsymbol{x}$$

$$\text{subject to} \quad \boldsymbol{A}\,\boldsymbol{x} = \boldsymbol{b}$$

$$\boldsymbol{x} \geq 0$$

- It is obtained from an LP problem in **canonical maximization form** with n non-negative variables and m less-than-or-equal-to linear constraints, by adding m slack non-negative variables

Recall: basis and dictionary

- Given a basis $\boldsymbol{A}_{\mathcal{B}}$, with basic indices $\mathcal{B}$ and non-basic indices $\mathcal{N}$, we use (as earlier in the course):

$$\bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b} \qquad \bar{\boldsymbol{A}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_{\mathcal{N}}$$

$$\bar{z} = \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b} \qquad \bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{c}'_{\mathcal{B}}\, \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_{\mathcal{N}}$$

- The associated dictionary is:

$$\begin{cases} x_i = \bar{b}_i - \sum_{j \in \mathcal{N}} \bar{a}_{ij}\, x_j, \quad \forall i \in \mathcal{B} \\ z = \bar{z} + \sum_{j \in \mathcal{N}} \bar{c}_j\, x_j \end{cases}$$

Sufficient conditions

Theorem 1: sufficient conditions for optimality

Given a basis $\boldsymbol{A}_{\mathcal{B}}$ *of the LP problem in standard form, if* $\bar{\boldsymbol{b}} \geq 0$ *and* $\bar{\boldsymbol{c}} \leq 0$ *then*

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \quad \text{and} \quad \boldsymbol{x}_{\mathcal{N}} = 0$$

is an optimal solution and $\boldsymbol{c}'_{\mathcal{B}}\, \bar{\boldsymbol{b}}$ *is the optimal solution value.*

How to prove it

- $\bar{\boldsymbol{b}} \geq 0$ gives **feasibility** of the basic solution
- $\bar{\boldsymbol{c}} \leq 0$ gives **optimality**: we write the objective function of any feasible solution through the basis and compare it with the value of the basic solution

Proof (1/2)

- Given a basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard form, if $\bar{\boldsymbol{b}} \geq 0$ then $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$ is a feasible solution
- If $\bar{\boldsymbol{c}} \leq 0$, for all feasible solutions $\boldsymbol{x}$ we have:

$$\boldsymbol{c}' \, \boldsymbol{x} = \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{x}_{\mathcal{B}} + \boldsymbol{c}'_{\mathcal{N}} \, \boldsymbol{x}_{\mathcal{N}} = \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{b} + \underbrace{\bar{\boldsymbol{c}}'}_{\leq 0} \, \underbrace{\boldsymbol{x}_{\mathcal{N}}}_{\geq 0}$$

$$\leq \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{b} = \boldsymbol{c}'_{\mathcal{B}} \, \bar{\boldsymbol{b}} + \boldsymbol{c}'_{\mathcal{N}} \, 0$$

Proof (2/2)

- Then the objective function value of all feasible solutions is smaller or equal to the objective function value of the basic feasible solution

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \quad \text{and} \quad \boldsymbol{x}_{\mathcal{N}} = 0$$

- Accordingly, the latter is an optimal solution of value $\boldsymbol{c}'_{\mathcal{B}}\, \bar{\boldsymbol{b}}$

□

A sufficient condition (1/2)

- We have proven that:

$$\bar{\boldsymbol{b}} \geq 0 \text{ and } \bar{\boldsymbol{c}} \leq 0 \implies \boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \text{ and } \boldsymbol{x}_{\mathcal{N}} = 0, \text{ optimal solution}$$

- So "$\bar{\boldsymbol{b}} \geq 0$ and $\bar{\boldsymbol{c}} \leq 0$" is a **sufficient condition** to "$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$, optimal solution"
- and "$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$, optimal solution" is a **necessary condition** to "$\bar{\boldsymbol{b}} \geq 0$ and $\bar{\boldsymbol{c}} \leq 0$"

A sufficient condition (2/2)

- The implication does **not** work in the other direction (we will see a counterexample)
- Accordingly:

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \text{ and } \boldsymbol{x}_{\mathcal{N}} = 0, \text{ optimal solution} \quad \not\Rightarrow \quad \bar{\boldsymbol{c}} \leq 0$$

When is the optimal solution unique?

- If the reduced costs are not only non-positive but **strictly negative**, we can say more
- Every move away from the basic solution then strictly decreases the objective function

Unique optimal solution

Corollary 1

Given a basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard form, if $\bar{\boldsymbol{b}} \geq 0$ and $\bar{\boldsymbol{c}} < 0$ then

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \quad \text{and} \quad \boldsymbol{x}_{\mathcal{N}} = 0$$

is the unique optimal solution.

Proof (1/2)

- It follows from the previous Theorem that $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$ is an optimal solution
- If $\bar{\boldsymbol{c}} < 0$ and since any feasible solution $\boldsymbol{x}$ different from $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$ must have $\boldsymbol{x}_{\mathcal{N}} \neq 0$, we have:

$$\boldsymbol{c}'\,\boldsymbol{x} = \boldsymbol{c}'_{\mathcal{B}}\,\boldsymbol{x}_{\mathcal{B}} + \boldsymbol{c}'_{\mathcal{N}}\,\boldsymbol{x}_{\mathcal{N}} = \boldsymbol{c}'_{\mathcal{B}}\,\boldsymbol{A}_{\mathcal{B}}^{-1}\,\boldsymbol{b} + \underbrace{\bar{\boldsymbol{c}}'}_{<0}\ \underbrace{\boldsymbol{x}_{\mathcal{N}}}_{\geq 0 \text{ and } \neq 0}$$

$$< \boldsymbol{c}'_{\mathcal{B}}\,\boldsymbol{A}_{\mathcal{B}}^{-1}\,\boldsymbol{b} = \boldsymbol{c}'_{\mathcal{B}}\,\bar{\boldsymbol{b}} + \boldsymbol{c}'_{\mathcal{N}}\,0$$

Proof (2/2)

- Every feasible solution different from the basic one has a strictly smaller objective function value
- Accordingly, $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$ is the unique optimal solution

□

A sufficient condition for uniqueness

- Accordingly, we have shown that:

$\bar{\boldsymbol{b}} \geq 0$ and $\bar{\boldsymbol{c}} < 0$

$$\Longrightarrow \quad \boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \text{ and } \boldsymbol{x}_{\mathcal{N}} = 0, \text{ unique optimal solution}$$

- So "$\bar{\boldsymbol{b}} \geq 0$ and $\bar{\boldsymbol{c}} < 0$" is a **sufficient condition** to "$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$, unique optimal solution"
- The implication does **not** work in the other direction (we will give a counterexample). Accordingly:

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \text{ and } \boldsymbol{x}_{\mathcal{N}} = 0, \text{ unique optimal solution}$$

$$\not\Rightarrow \quad \bar{\boldsymbol{c}} < 0$$

We now see these conditions at work
on a small LP problem with 3 variables and 2 constraints,

and we find the promised **counterexamples**.

Example: standard max form (1/5)

- We introduce the following LP problem in **canonical max form** with $n = 3$ variables and $m = 2$ constraints:

$$
\begin{array}{lrcrcrcl}
\max & x_1 & + & 2\,x_2 & + & 7\,x_3 & & \\
\text{subject to} & x_1 & & & + & 2\,x_3 & \leq & 2 \\
 & & & x_2 & + & 2\,x_3 & \leq & 2 \\
 & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$

Example: standard max form (2/5)

- Its **standard form** with $n + m = 5$ variables and $m = 2$ constraints is:

$$
\begin{array}{lccccccccc}
\max & x_1 & + & 2\,x_2 & + & 7\,x_3 & & & & \\
\text{subject to} & x_1 & & & + & 2\,x_3 & + x_4 & & = & 2 \\
 & & & x_2 & + & 2\,x_3 & & + x_5 & = & 2 \\
 & x_1, & & x_2, & & x_3, & x_4, & x_5 & \geq & 0
\end{array}
$$

- x_4 and x_5 are the slack variables of the two constraints

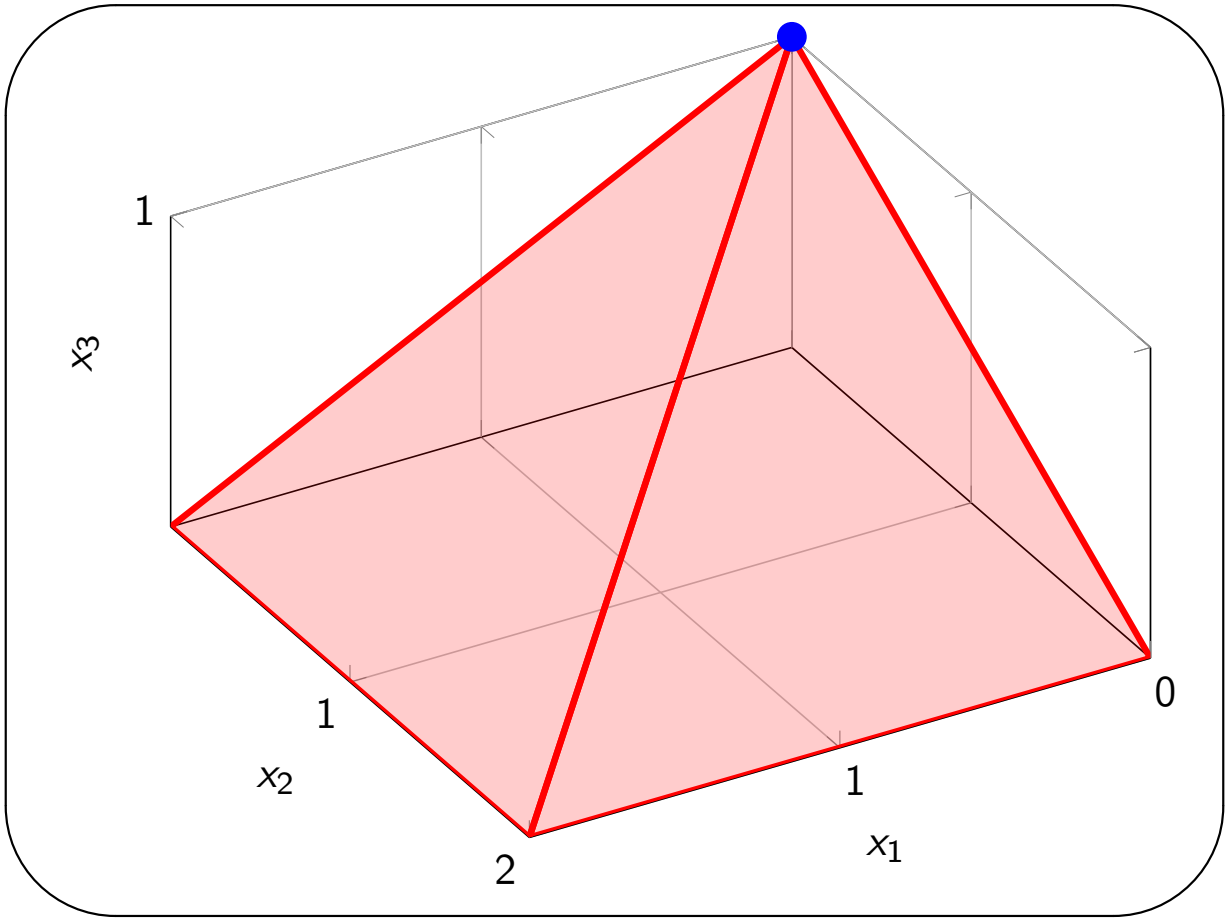
Example: standard max form (3/5)

- In the LP problem in standard form, we have:

$$\boldsymbol{c} = \begin{pmatrix} 1 \\ 2 \\ 7 \\ 0 \\ 0 \end{pmatrix} \quad \boldsymbol{A} = \begin{pmatrix} 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1 \end{pmatrix} \quad \boldsymbol{b} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \quad \text{and} \quad \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix}$$

Example: standard max form (4/5)

- The feasible region of the canonical form and its optimal vertex (in blue):



Example: standard max form (5/5)

- An optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}\,)$ of the canonical form is:

$$\tilde{x}_1 = 0, \quad \tilde{x}_2 = 0, \quad \tilde{x}_3 = 1$$

and the optimal solution value is:

$$z(\,\mathrm{LP}\,) = 7$$

- An optimal solution $\tilde{\boldsymbol{x}}(\,\mathrm{LP}\,)$ of the standard form is:

$$\tilde{x}_1 = 0, \quad \tilde{x}_2 = 0, \quad \tilde{x}_3 = 1, \quad \tilde{x}_4 = 0, \quad \tilde{x}_5 = 0$$

with the same optimal solution value

Example: optimality conditions with $\mathcal{B} = \{2, 3\}$ (1/7)

- For the LP problem of the previous example, with $\mathcal{B} = \{2, \ 3\}$ and $\mathcal{N} = \{1, \ 4, \ 5\}$, we have:

$$\boldsymbol{x}_{\mathcal{B}} = \begin{pmatrix} x_2 \\ x_3 \end{pmatrix} \quad \boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} 0 & 2 \\ 1 & 2 \end{pmatrix} \quad \boldsymbol{A}_{\mathcal{B}}^{-1} = \begin{pmatrix} -1 & 1 \\ \frac{1}{2} & 0 \end{pmatrix} \quad \boldsymbol{c}_{\mathcal{B}} = \begin{pmatrix} 2 \\ 7 \end{pmatrix}$$

$$\boldsymbol{x}_{\mathcal{N}} = \begin{pmatrix} x_1 \\ x_4 \\ x_5 \end{pmatrix} \quad \boldsymbol{A}_{\mathcal{N}} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \boldsymbol{c}_{\mathcal{N}} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

Example: optimality conditions with $\mathcal{B} = \{2, 3\}$ (2/7)

- Accordingly, we have:

$$\bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b} = \begin{pmatrix} -1 & 1 \\ \frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$\bar{\boldsymbol{A}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_{\mathcal{N}} = \begin{pmatrix} -1 & 1 \\ \frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & -1 & 1 \\ \frac{1}{2} & \frac{1}{2} & 0 \end{pmatrix}$$

Example: optimality conditions with $\mathcal{B} = \{2, 3\}$ (3/7)

- The value of the basic solution is:

$$\bar{z} = \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{b} = \begin{pmatrix} 2 & 7 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ \frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{3}{2} & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 7$$

Example: optimality conditions with $\mathcal{B} = \{2, 3\}$ (4/7)

- The reduced costs are:

$$\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{A}_{\mathcal{N}}$$

$$= \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 2 & 7 \end{pmatrix} \begin{pmatrix} -1 & 1 \\ \frac{1}{2} & 0 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} - \begin{pmatrix} \frac{3}{2} & 2 \end{pmatrix} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \end{pmatrix} - \begin{pmatrix} \frac{3}{2} & \frac{3}{2} & 2 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & -\frac{3}{2} & -2 \end{pmatrix}$$

Example: optimality conditions with $\mathcal{B} = \{2, 3\}$ (5/7)

- Since $\bar{\boldsymbol{b}} \geq 0$ and $\bar{\boldsymbol{c}} \leq 0$, the basic feasible solution

$$\begin{pmatrix} x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x_1 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

  is the **unique optimal solution** and $\bar{z} = 7$ is the optimal solution value

- Indeed, all the reduced costs are strictly negative: $\bar{\boldsymbol{c}} < 0$

Example: optimality conditions with $\mathcal{B} = \{2, 3\}$ (6/7)

- The associated dictionary is:

$$\begin{cases} x_2 = 0 + x_1 + x_4 - x_5 \\ x_3 = 1 - \frac{1}{2} x_1 - \frac{1}{2} x_4 \\ z = 7 - \frac{1}{2} x_1 - \frac{3}{2} x_4 - 2 x_5 \end{cases}$$

Example: optimality conditions with $\mathcal{B} = \{2, 3\}$ (7/7)

- In the dictionary we read directly the two conditions:
  - the constant terms of the rows of x_2 and x_3 are $\bar{\boldsymbol{b}} = (0,\ 1)' \geq 0$
  - the coefficients of the row of z are $\bar{\boldsymbol{c}}' = (-\frac{1}{2},\ -\frac{3}{2},\ -2) < 0$
- Note that the basic variable x_2 takes the value 0

Same LP problem, same optimal vertex,
but a **different basis**:

what do the reduced costs say now?

Example: optimality conditions with $\mathcal{B} = \{3, 5\}$ (1/7)

- For the LP problem of the same example, with $\mathcal{B} = \{3, \ 5\}$ and $\mathcal{N} = \{1, \ 2, \ 4\}$, we have:

$$\boldsymbol{x}_{\mathcal{B}} = \begin{pmatrix} x_3 \\ x_5 \end{pmatrix} \quad \boldsymbol{A}_{\mathcal{B}} = \begin{pmatrix} 2 & 0 \\ 2 & 1 \end{pmatrix} \quad \boldsymbol{A}_{\mathcal{B}}^{-1} = \begin{pmatrix} \frac{1}{2} & 0 \\ -1 & 1 \end{pmatrix} \quad \boldsymbol{c}_{\mathcal{B}} = \begin{pmatrix} 7 \\ 0 \end{pmatrix}$$

$$\boldsymbol{x}_{\mathcal{N}} = \begin{pmatrix} x_1 \\ x_2 \\ x_4 \end{pmatrix} \quad \boldsymbol{A}_{\mathcal{N}} = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad \boldsymbol{c}_{\mathcal{N}} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}$$

Example: optimality conditions with $\mathcal{B} = \{3, 5\}$ (2/7)

- Accordingly, we have:

$$\bar{\boldsymbol{b}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{b} = \begin{pmatrix} \frac{1}{2} & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\bar{\boldsymbol{A}} = \boldsymbol{A}_{\mathcal{B}}^{-1}\, \boldsymbol{A}_{\mathcal{N}} = \begin{pmatrix} \frac{1}{2} & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & 0 & \frac{1}{2} \\ -1 & 1 & -1 \end{pmatrix}$$

Example: optimality conditions with $\mathcal{B} = \{3, 5\}$ (3/7)

- The value of the basic solution is:

$$\bar{z} = \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{b} = \begin{pmatrix} 7 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{7}{2} & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 7$$

Example: optimality conditions with $\mathcal{B} = \{3, 5\}$ (4/7)

- The reduced costs are:

$$\bar{\boldsymbol{c}}' = \boldsymbol{c}'_{\mathcal{N}} - \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{A}_{\mathcal{N}}$$

$$= \begin{pmatrix} 1 & 2 & 0 \end{pmatrix} - \begin{pmatrix} 7 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & 0 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 0 \end{pmatrix} - \begin{pmatrix} \frac{7}{2} & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 0 \end{pmatrix} - \begin{pmatrix} \frac{7}{2} & 0 & \frac{7}{2} \end{pmatrix} = \begin{pmatrix} -\frac{5}{2} & 2 & -\frac{7}{2} \end{pmatrix}$$

Example: optimality conditions with $\mathcal{B} = \{3, 5\}$ (5/7)

- Since $\bar{\boldsymbol{b}} \geq 0$ and $\bar{z} = 7$ (the optimal solution value), the basic feasible solution

$$\begin{pmatrix} x_3 \\ x_5 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x_1 \\ x_2 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

  is an **optimal solution** but $\bar{\boldsymbol{c}} \not\geq 0$ (the reduced cost of x_2 is $2 > 0$)

Example: optimality conditions with $\mathcal{B} = \{3, 5\}$ (6/7)

- It is a **counterexample** to the fact that a basic optimal solution implies non-positive reduced costs
- Since this optimal solution is unique (see the example with $\mathcal{B} = \{2, 3\}$), it is also a **counterexample** to the fact that a unique basic optimal solution implies negative reduced costs

Example: optimality conditions with $\mathcal{B} = \{3, 5\}$ (7/7)

- The associated dictionary is:

$$\begin{cases} x_3 = 1 - \frac{1}{2}x_1 - \frac{1}{2}x_4 \\ x_5 = 0 + x_1 - x_2 + x_4 \\ z = 7 - \frac{5}{2}x_1 + 2x_2 - \frac{7}{2}x_4 \end{cases}$$

- The basic variable x_5 takes the value 0: this is what allows an optimal basic solution with a positive reduced cost, as we see in the next section

Summary

Sufficient conditions for optimality:

- $\bar{\boldsymbol{b}} \geq 0$ and $\bar{\boldsymbol{c}} \leq 0 \Longrightarrow \boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$, $\boldsymbol{x}_{\mathcal{N}} = 0$ is an optimal solution, of value $\boldsymbol{c}'_{\mathcal{B}}\, \bar{\boldsymbol{b}}$
- $\bar{\boldsymbol{b}} \geq 0$ and $\bar{\boldsymbol{c}} < 0 \Longrightarrow$ it is the unique optimal solution

Not necessary:

- With $\mathcal{B} = \{3, 5\}$ the basic solution is the unique optimal solution, but $\bar{\boldsymbol{c}} \not\leq 0$

Necessary conditions for optimality

Introduction: Necessary conditions for optimality

- The counterexample of the previous section has a basic variable at zero
- We give a name to these solutions: **degenerate basic feasible solutions**
- We prove that, for a **non-degenerate** optimal solution, the reduced costs must be non-positive: the conditions become necessary and sufficient
- We also define the **degenerate pivoting operations**, in which the objective function can remain unchanged

Degenerate basic feasible solutions

Definition 1: degenerate basic feasible solutions

Given a basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard form, if $\bar{\boldsymbol{b}} \geq 0$ and for some $i \in \mathcal{B}$ we have $\bar{b}_i = 0$ then

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \quad \text{and} \quad \boldsymbol{x}_{\mathcal{N}} = 0$$

*is called a **degenerate basic feasible solution**.*

- In a degenerate basic feasible solution there are **more than** n **variables at zero** (also some basic variables are at zero and not only all the non-basic variables)

In a non-degenerate solution all the basic variables are **strictly positive**:

there is room to increase a non-basic variable.

This is the key of the necessary conditions.

Necessary conditions

Theorem 2: necessary conditions for optimality

Given a basis $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard form, if

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} \quad \textit{and} \quad \boldsymbol{x}_{\mathcal{N}} = 0$$

is a non degenerate optimal solution then $\bar{\boldsymbol{c}} \leq 0$.

Proof (1/2)

- Suppose by contradiction that $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$ is a non degenerate optimal solution and that $\bar{c}_p > 0$ for some $p \in \mathcal{N}$
- Since the solution is non degenerate and feasible we have $\bar{\boldsymbol{b}} > 0$
- But then there is a positive value $\vartheta > 0$ such that:

$$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} - \underbrace{\boldsymbol{A}_{\mathcal{B}}^{-1} \boldsymbol{A}_p}_{=\bar{\boldsymbol{A}}_p} \vartheta \geq 0$$

Proof (2/2)

- Then by setting

$$x_p = \vartheta \quad \text{and} \quad \boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}} - \bar{\boldsymbol{A}}_p \, \vartheta$$

  (all other variables at zero), we get another feasible solution with a strictly better objective function value

- This is a contradiction with $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$ being optimal

- So we must have $\bar{\boldsymbol{c}} \leq 0$

□

Necessary and sufficient conditions

- Combined with the sufficient conditions for optimality, we have shown that:
  $\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$, non degenerate optimal solution

  $$\iff \quad \bar{\boldsymbol{b}} > 0 \text{ and } \bar{\boldsymbol{c}} \leq 0$$

- So "$\bar{\boldsymbol{b}} > 0$ and $\bar{\boldsymbol{c}} \leq 0$" is a **necessary and sufficient condition** to "$\boldsymbol{x}_{\mathcal{B}} = \bar{\boldsymbol{b}}$ and $\boldsymbol{x}_{\mathcal{N}} = 0$, non-degenerate optimal solution" and vice-versa

Example: degenerate basic feasible solutions

- For the LP problem of the previous section, with $\mathcal{B} = \{3, \ 5\}$ and $\mathcal{N} = \{1, \ 2, \ 4\}$, we have the **degenerate** basic feasible solution:

$$\begin{pmatrix} x_3 \\ x_5 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x_1 \\ x_2 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

- The basic variable x_5 is at zero: the theorem does not apply, and indeed the reduced cost of x_2 is positive

From solutions to pivoting operations

- Degeneracy also affects the **pivoting operations** of the simplex algorithm
- In a pivoting operation, the pivot column is increased to its maximum possible value t_m, and the pivot row x_r is a basic variable that reaches zero
- What happens when $t_m = 0$?

Degenerate pivoting operations

Definition 2: degenerate pivoting operation

*A **degenerate pivoting operation** occurs when $t_m = 0$, i.e., when the pivot row x_r has $\bar{b}_r = 0$; this can happen only if the basic feasible solution is degenerate.*

The objective function in the pivoting operations

The **value of the objective function increases in each non degenerate pivoting operation**

and **can remain unchanged only in a degenerate one**.

Summary

Degenerate basic feasible solutions:

- $\bar{\boldsymbol{b}} \geq 0$ and $\bar{b}_i = 0$ for some $i \in \mathcal{B}$: more than n variables at zero

Necessary conditions:

- Non degenerate optimal solution $\Longrightarrow \bar{\boldsymbol{c}} \leq 0$
- Non degenerate optimal solution $\Longleftrightarrow \bar{\boldsymbol{b}} > 0$ and $\bar{\boldsymbol{c}} \leq 0$

Degenerate pivoting operations:

- $t_m = 0$: the objective function value can remain unchanged only in these operations

Pivot rules

Introduction: Pivot rules

- In a pivoting operation there may be **several candidates** for the pivot column and for the pivot row: a pivot rule decides
- With degenerate pivoting operations the objective function may not increase: the simplex algorithm could **cycle** forever
- We prove that a non-terminating simplex algorithm must cycle, and that **Bland's pivot rule** prevents cycling
- With Bland's rule the simplex algorithm always terminates

What is a pivot rule

- In pivoting operations, a **pivot rule** is a way of answering the two questions:

1. How to determine the **pivot column** in cases where there are multiple non-basic variables with positive reduced costs?

2. How to determine the **pivot row** in cases where there are multiple basic variables taking a value of zero when setting the value of the pivot column to its maximum possible value t_m?

Termination and cycling

Theorem 3

If the simplex algorithm fails to terminate, then it must cycle.

Proof

- There are at most
$$\binom{n+m}{m}$$
bases $\boldsymbol{A}_{\mathcal{B}}$ of the LP problem in standard form and, accordingly, dictionaries

- Then, if the simplex algorithm fails to terminate, it must visit some of these dictionaries more than once

- Hence, the algorithm cycles

□

Cycling in practice

- **It is very rare that the simplex algorithm cycles, but instances in which it cycles exist**
- Cycling can occur only within a sequence of **degenerate basic feasible solutions** when the objective function remains unchanged
- An anti-cycling pivot rule chooses the pivot column and the pivot row so that this can never happen

Bland's pivot rule

- **Bland's pivot rule** (smallest-index pivot rule) selects the pivot column and the pivot row as follows:

1. the **pivot column** is the non-basic variable x_p with the **smallest index** $p \in \mathcal{N}$ such that:

$$\bar{c}_p > 0 \quad \text{or equivalently} \quad c_p > \boldsymbol{c}'_{\mathcal{B}} \, \boldsymbol{A}_{\mathcal{B}}^{-1} \, \boldsymbol{A}_p$$

2. the **pivot row** is the basic variable x_r with the **smallest index** $r \in \mathcal{B}$ such that:

$$\bar{a}_{rp} > 0 \quad \text{and} \quad \frac{\bar{b}_r}{\bar{a}_{rp}} = t_m$$

Bland's rule terminates

Theorem 4

The simplex algorithm terminates as long as the pivot column and pivot row are selected by Bland's pivot rule in each pivoting operation.

- According to the previous Theorem, the proof only needs to show that **cycling is impossible** when Bland's pivot rule is used

The idea of the proof

- We assume that a cycle exists and look at the variables that enter and leave the basis along the cycle
- We compare two dictionaries of the cycle: one where the variable with the largest index leaves the basis, and one where it enters
- Bland's rule then forces a different choice of the pivot row: a contradiction

Proof (1/10)

- We proceed by contradiction and we assume that cycling does occur
- Accordingly, let

$$D_0, \quad D_1, \quad D_2, \quad \ldots \quad , D_{k-1}, \quad D_0, \quad D_1, \quad D_2, \quad \ldots$$

be a sequence of dictionaries such that the dictionary D_0 is repeated after k degenerate pivoting operations with constant objective function value

- We say that a variable is ***fickle*** if it is a basic variable in some dictionaries and a non-basic variable in some others

Proof (2/10)

- Let x_t be the fickle variable having the **largest index** t and let D denote a dictionary

$$\begin{cases} x_i = \bar{b}_i - \sum_{j \in \mathcal{N}} \bar{a}_{ij}\, x_j, \quad \forall i \in \mathcal{B} \\ z = \bar{z} + \sum_{j \in \mathcal{N}} \bar{c}_j\, x_j \end{cases}$$

  of the cycling sequence $D_0, \quad D_1, \quad D_2, \quad \ldots \quad , D_{k-1}$ in which x_t is the **pivot row**

- Let x_s denote the **pivot column** of the dictionary D
- W.l.o.g., we assume that $D = D_0$

Proof (3/10)

- Let D^* denote a dictionary

$$\begin{cases} x_i = \bar{b}_i^* - \sum_{j \in \mathcal{N}^*} \bar{a}_{ij}^* \, x_j, \quad \forall i \in \mathcal{B}^* \\ z = \bar{z}^* + \sum_{j \in \mathcal{N}^*} \bar{c}_j^* \, x_j \end{cases}$$

  of the sequence $D_1, \quad D_2, \quad \ldots \quad , D_{k-1}$ in which x_t is the **pivot column**

- Such a dictionary exists: x_t leaves the basis in D and, since D_0 is repeated, it must enter it again

Proof (4/10)

- Since all the pivoting operations in the cycle are degenerate (the objective function value remains constant), we have that $\bar{z} = \bar{z}^* = v$
- Therefore we can write the objective function of the dictionary D^* as

$$z = v + \sum_{j=1}^{n+m} \hat{c}_j^* \, x_j \quad \text{where} \quad \hat{c}_j^* = \begin{cases} \bar{c}_j^* & \text{if } \; j \in \mathcal{N}^* \\ 0 & \text{otherwise} \end{cases}$$

Proof (5/10)

- We consider the solution obtained by increasing the value of x_s to ϑ:

$$x_s = \vartheta, \quad x_i = 0, \ \forall i \in \mathcal{N} \setminus \{s\}, \quad x_i = \bar{b}_i - \bar{a}_{is}\,\vartheta, \ \forall i \in \mathcal{B}$$

- Accordingly, for the objective function z of the dictionaries D and D^*, we have:

$$z = v + \bar{c}_s\,\vartheta \quad \text{and} \quad z = v + \hat{c}^*_s\,\vartheta + \sum_{i \in \mathcal{B}} \hat{c}^*_i\,\left(\bar{b}_i - \bar{a}_{is}\vartheta\right)$$

Proof (6/10)

- Equating these two expressions for z, we have:

$$\left(\bar{c}_s - \hat{c}_s^* + \sum_{i \in \mathcal{B}} \hat{c}_i^* \, \bar{a}_{is} \right) \vartheta = \sum_{i \in \mathcal{B}} \hat{c}_i^* \, \bar{b}_i$$

$$\Rightarrow \quad \bar{c}_s - \hat{c}_s^* + \sum_{i \in \mathcal{B}} \hat{c}_i^* \, \bar{a}_{is} = 0$$

- The implication is due to the fact that the equation must hold for every ϑ (accordingly, we also have $\sum_{i \in \mathcal{B}} \hat{c}_i^* \, \bar{b}_i = 0$)

Proof (7/10)

- The fact that x_s is the pivot column in D implies that $\bar{c}_s > 0$
- Since x_s is fickle and $s \neq t$ (x_s is non-basic and x_t is basic in D), we have $s < t$
- Moreover, we have $\hat{c}^*_s \leq 0$:
  - if $s \in \mathcal{B}^*$, then $\hat{c}^*_s = 0$;
  - if $s \in \mathcal{N}^*$ and $\hat{c}^*_s = \bar{c}^*_s > 0$, then, according to Bland's rule and since $s < t$, x_s and not x_t would be the pivot column in D^*

Proof (8/10)

- Accordingly, we have:

$$\sum_{i \in \mathcal{B}} \hat{c}_i^* \, \bar{a}_{is} = \hat{c}_s^* - \bar{c}_s < 0$$

$$\Rightarrow \quad \text{there exists } r \in \mathcal{B} : \ \hat{c}_r^* \bar{a}_{rs} < 0$$

$$\text{and then } \ \hat{c}_r^* \neq 0 \text{ and } r \in \mathcal{N}^*$$

- Hence x_r is fickle and therefore $r \leq t$
- In fact, $r < t$, since $\hat{c}_t^* \bar{a}_{ts} > 0$: both factors are positive, since
  - $\hat{c}_t^* = \bar{c}_t^* > 0$ (x_t is the pivot column in D^*)
  - $\bar{a}_{ts} > 0$ (x_t is the pivot row in D)

Proof (9/10)

- The fact that $r < t$ implies that $\hat{c}_r^* \leq 0$ (otherwise, according to Bland's rule, x_r would be the pivot column in D^*)
- Then we have:

$$\hat{c}_r^* \bar{a}_{rs} < 0 \quad \text{and} \quad \hat{c}_r^* \leq 0 \quad \Rightarrow \quad \bar{a}_{rs} > 0$$

- Since each of the dictionaries in the cycle describes the same feasible solution (all pivoting operations are degenerate), every fickle variable is zero in all these dictionaries (it is zero in a dictionary in which it is non-basic)

Proof (10/10)

- In particular, $x_r = 0$. But in D, x_r is basic
- Hence $\bar{b}_r = 0$ and, since $\bar{a}_{rs} > 0$, we have

$$\frac{\bar{b}_r}{\bar{a}_{rs}} = 0 = t_m$$

- So x_r was a candidate to be the pivot row in D, and since $r < t$, according to Bland's rule it should have been chosen instead of x_t
- This is the contradiction that proves that a cycling sequence of dictionaries cannot exist using Bland's rule

□

Summary

Pivot rules:

- They choose the pivot column among the non-basic variables with positive reduced cost, and the pivot row among the basic variables that reach zero at t_m

Cycling:

- A simplex algorithm that fails to terminate must cycle (at most $\binom{n+m}{m}$ bases)
- Cycling is very rare and can occur only along degenerate pivoting operations

Bland's pivot rule:

- Smallest index for the pivot column and for the pivot row
- With Bland's rule the simplex algorithm always terminates

Summary (1/2)

Sufficient conditions for optimality:

- $\bar{\boldsymbol{b}} \geq 0$ and $\bar{\boldsymbol{c}} \leq 0$: the basic solution is optimal, of value $\boldsymbol{c}'_{\mathcal{B}}\, \bar{\boldsymbol{b}}$
- $\bar{\boldsymbol{b}} \geq 0$ and $\bar{\boldsymbol{c}} < 0$: the basic solution is the unique optimal solution
- The converse implications do not hold (counterexample with $\mathcal{B} = \{3, 5\}$)

Necessary conditions for optimality:

- A non degenerate optimal basic solution has $\bar{\boldsymbol{c}} \leq 0$
- Non degenerate optimal solution $\iff \bar{\boldsymbol{b}} > 0$ and $\bar{\boldsymbol{c}} \leq 0$

Summary (2/2)

Degeneracy:

- A degenerate basic feasible solution has some basic variables at zero
- In a degenerate pivoting operation $t_m = 0$ and the objective function can remain unchanged

Pivot rules:

- If the simplex algorithm fails to terminate, then it must cycle
- Bland's pivot rule (smallest index) guarantees that the simplex algorithm terminates